- **Module-2(8 hours)**

- Operational amplifiers -Operational amplifier parameters,
- Operational amplifier characteristics,
- Operational amplifier configurations,
- Operational amplifier circuits.

- Oscillators – Barkhausen criterion,
- sinusoidal and non-sinusoidal oscillators,
- Ladder network oscillator,
- Wein bridge oscillator,
- Multivibrators, Single-stage astable oscillator,
- Crystal controlled oscillators

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Operational Amplifier:

- **Operational Amplifiers**, or **Op-amps** as they are more commonly called, are one of the basic building blocks of Analog Electronic Circuits.
- Operational amplifiers are linear devices which have all the properties required for nearly ideal DC amplification.
- They are used extensively in signal conditioning, filtering or to perform mathematical operations such as add, subtract, integration and differentiation.
- An Operational Amplifier is basically a three-terminal device which consists of two high impedance inputs.
- One of the inputs is called the **Inverting Input**, marked with a negative or “minus” sign, (–). The other input is called the **Non-inverting Input**, marked with a positive or “plus” sign (+).
- A third terminal represents the operational amplifiers output port which can both sink and source either a voltage or a current.

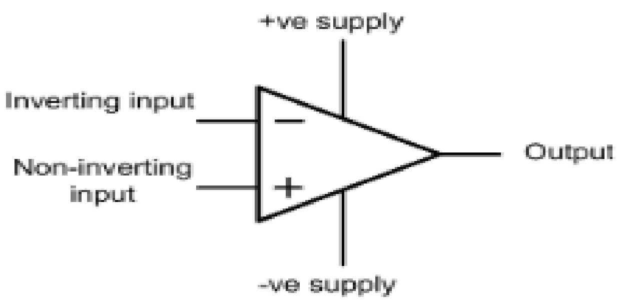


FIG.2.1. Symbol for an operational amplifier.

Operational amplifier parameters:

The important parameters of op-amps are

1) Open-loop voltage gain

- The open-loop voltage gain of an operational amplifier is defined as the ratio of output voltage to input voltage measured with no feedback applied.
- In practice, this value is exceptionally high (typically greater than 100,000) but is liable to considerable variation from one device to another.

Open-loop voltage gain may thus be thought of as the 'internal' voltage gain of the device, thus:

$$A_{V(OL)} = \frac{V_{OUT}}{V_{IN}}$$

Where $A_{V(OL)}$ is the open-loop voltage gain, V_{OUT} and V_{IN} are the output and input voltages, respectively, under open-loop conditions.

The open-loop voltage gain is often expressed in decibels (dB) rather than as a ratio.

In this case:

$$A_{V(OL)} = 20\log_{10}\frac{V_{OUT}}{V_{IN}}$$

Most operational amplifiers have open-loop voltage gains of 90 dB or more.

2) Closed-loop voltage gain:

- The closed-loop voltage gain of an operational amplifier is defined as the ratio of output voltage to input voltage measured with a small proportion of the output fed-back to the input (i.e. with feedback applied).
- The effect of providing negative feedback is to reduce the loop voltage gain to a value that is both predictable and manageable.
- Practical closed-loop voltage gains range from one to several thousand but note that high values of voltage gain may make unacceptable restrictions on bandwidth.
- Closed-loop voltage gain is once again the ratio of output voltage to input voltage but with negative feedback applied, hence:

$$A_{V(CL)} = \frac{V_{OUT}}{V_{IN}}$$

- Where $A_{V(CL)}$ is the open-loop voltage gain, V_{OUT} and V_{IN} are the output and input voltages, respectively, under closed-loop conditions.
- The closed-loop voltage gain is normally very much less than the open-loop voltage gain.

Problem :

An operational amplifier operating with negative feedback produces an output voltage of 2 V when supplied with an input of 400 μV. Determine the value of closed-loop voltage gain.

$$A_{V(CL)} = \frac{V_{OUT}}{V_{IN}}$$

Thus:

$$A_{V(CL)} = \frac{2}{400 \times 10^{-6}} = \frac{2 \times 10^{6}}{400} = 5{,}000$$

Expressed in decibels (rather than as a ratio) this is:

$$A_{V(CL)} = 20 \log_{10}(5{,}000) = 20 \times 3.7 = 74 \text{ dB}$$

3) Input resistance:

- The input resistance of an operational amplifier is defined as the ratio of input voltage to input current expressed in ohms.
- It is often expedient to assume that the input of an operational amplifier is purely resistive, though this is not the case at high frequencies where shunt capacitive reactance may become significant.
- The input resistance of operational amplifiers is very much dependent on the semiconductor technology employed.
- In practice values range from about 2 MΩ to over 10^{12} Ω.
- Input resistance is the ratio of input voltage to input current:

$$R_{IN} = \frac{V_{IN}}{I_{IN}}$$

- Where R_{IN} is the input resistance (in ohms), V_{IN} is the input voltage (in volts) and I_{IN} is the input current (in amps).

Problem :

An operational amplifier has an input resistance of 2 MΩ. Determine the input current when an input voltage of 5 mV is present.

Solution

Now:

$$R_{IN} = \frac{V_{IN}}{I_{IN}}$$

thus

$$I_{IN} = \frac{V_{IN}}{R_{IN}} = \frac{5 \times 10^{-3}}{2 \times 10^{6}} = 2.5 \times 10^{-9} \text{A} = 2.5 \text{ nA}$$

4) Output resistance:

- The output resistance of an operational amplifier is defined as the ratio of open-circuit output voltage to short-circuit output current expressed in ohms.
- Typical values of output resistance range from less than 10 Ω to around 100 Ω, depending upon the configuration and amount of feedback employed.
- Output resistance is the ratio of open-circuit output voltage to short-circuit output current, hence:

$$R_{OUT} = \frac{V_{OUT(OC)}}{I_{OUT(SC)}}$$

- where R_{OUT} is the output resistance (in ohms), $V_{OUT(OC)}$ is the open-circuit output voltage (in volts) and $I_{OUT(SC)}$ is the short-circuit output current (in amps).

5) Input offset voltage

- An ideal operational amplifier would provide zero output voltage when 0 V difference is applied to its inputs.
- In practice, due to imperfect internal balance, there may be some small voltage present at the output.
- The voltage that must be applied differentially to the operational amplifier input in order to make the output voltage exactly zero is known as the input offset voltage.
- Input offset voltage may be minimized by applying relatively large amounts of negative feedback or by using the offset null facility provided by a number of operational amplifier devices.

- Typical values of input offset voltage range from 1 mV to 15 mV.

6) Full-power bandwidth

- The full-power bandwidth for an operational amplifier is equivalent to the frequency at which the maximum undistorted peak output voltage swing falls to 0.707 of its low-frequency (d.c.) value (the sinusoidal input voltage remaining constant).
- Typical full-power bandwidths range from 10 kHz to over 1 MHz for some high-speed devices.

7) Slew rate

- Slew rate is the rate of change of output voltage with time, when a rectangular step input voltage is applied (as shown in Fig. 8.4).
- The slew rate of an operational amplifier is the rate of change of output voltage with time in response to a perfect step-function input. Hence:

$$\text{Slew rate} = \frac{\Delta V_{OUT}}{\Delta t}$$

- Where V_{OUT} is the change in output voltage (involts) and t is the corresponding interval of time (in seconds).
- Slew rate is measured in V/s (or V/µs) and typical values range from 0.2 V/µs to over 20 V/µs.

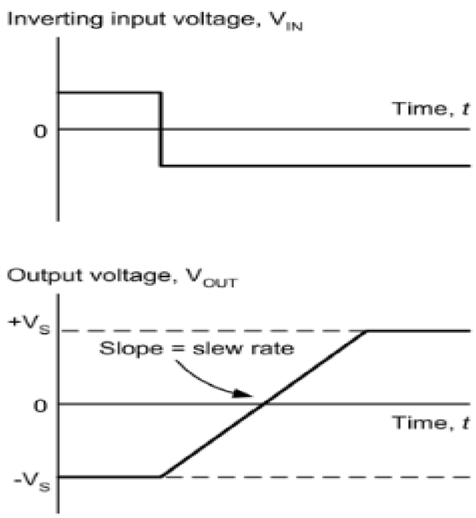


Fig 2.2. Slew rate for operational amplifier.

Operational amplifier characteristics:

- Consider the desirable characteristics for an 'ideal' operational amplifier. These are :
  (a) The open-loop voltage gain should be very high (ideally infinite).
  (b) The input resistance should be very high (ideally infinite).
  (c) The output resistance should be very low (ideally zero).
  (d) Full-power bandwidth should be as wide as possible.
  (e) Slew rate should be as large as possible.
  (f) Input offset should be as small as possible.
- The characteristics of most modern integrated circuit operational amplifiers (i.e. 'real' operational amplifiers) come very close to those of an 'ideal' operational amplifier, as witnessed by the data shown in Table 2.1

Table 2.1 Comparison of operational amplifier parameters for 'ideal' and 'real' devices.

Parameter	Ideal	Real
Voltage gain	Infinite	100,000
Input resistance	Infinite	100 MΩ
Output resistance	Zero	20 Ω
Bandwidth	Infinite	2 MHz
Slew rate	Infinite	10 V/μs
Input offset	Zero	Less than 5 mV

Problem:
A perfect rectangular pulse is applied to the input of an operational amplifier. If it takes 4 μs for the output voltage to change from –5 V to +5 V, determine the slew rate of the device.

Solution

The slew rate can be determined from:

$$\text{Slew rate} = \frac{\Delta V_{OUT}}{\Delta t} = \frac{10\ \text{V}}{4\ \mu\text{s}} = 2.5\ \text{V}/\mu\text{s}$$

Problem

A wideband operational amplifier has a slew rate of 15 V/μs. If the amplifier is used in a circuit with a voltage gain of 20 and a perfect step input of 100 mV is applied to its input, determine the time taken for the output to change level.

Solution

The output voltage change will be 20 × 100 = 2,000 mV (or 2 V). Re-arranging the formula for slew rate gives:

$$\Delta t = \frac{\Delta V_{OUT}}{\text{Slew rate}} = \frac{2\text{ V}}{15\text{ V}/\mu\text{s}} = 0.133\ \mu\text{s}$$

Operational amplifier configurations

- The three basic configurations for operational voltage amplifiers, together with the expressions for their voltage gain, are shown in Fig. 2.3.
- Supply rails have been omitted from these diagrams for clarity but are assumed to be symmetrical about 0 V.
- Signal may be superimposed on an unwanted d.c. voltage level or when the bandwidth of the amplifier greatly exceeds that of the signal that it is required to amplify.
- In such cases, capacitors of appropriate value may be inserted in series with the input resistor, R_{IN} , and in parallel with the feedback resistor, R_F , as shown in Fig. 8.8.
- The value of the input and feedback capacitors, C_{IN} and C_F respectively, are chosen so as to roll- off the frequency response of the amplifier at the desired lower and upper cut-off frequencies, respectively.
- The effect of these two capacitors on an operational amplifier's frequency response is shown in Fig. 8.9.
- By selecting appropriate values of capacitor, the frequency response of an inverting operational voltage amplifier may be very easily tailored to suit a particular set of requirements.
- The lower cut-off frequency is determined by the value of the input capacitance, C_{IN} , and input resistance, R_{IN} . The lower cut-off frequency is given by

$$f_1 = \frac{1}{2\pi C_{IN} R_{IN}} = \frac{0.159}{C_{IN} R_{IN}}$$

Where f1 is the lower cut-off frequency in hertz, C_{IN} is in farads and R_{IN} is in ohms.

Upper cut-off frequency will be determined by the feedback capacitance, C_F , and feedback resistance, R_F, such that:

$$f_2 = \frac{1}{2\pi C_F R_F} = \frac{0.159}{C_F R_F}$$

Where f_2 is the upper cut-off frequency in hertz, C_F is in farads and R_2 is in ohms.

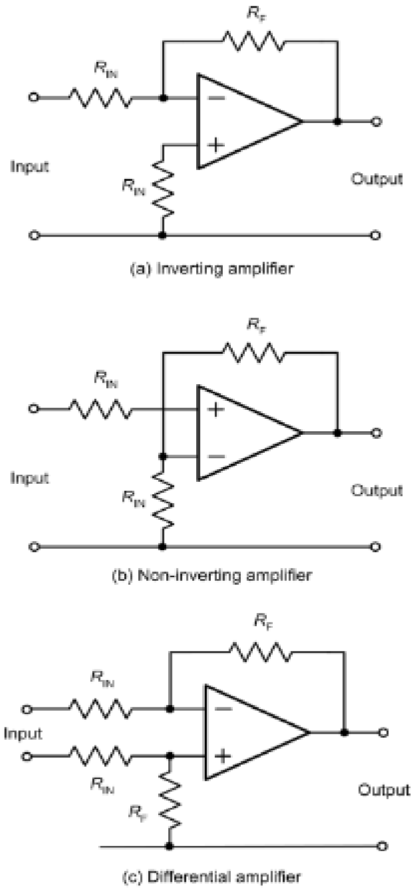


Fig 2.3. Three basic configuration for operational amplifier.

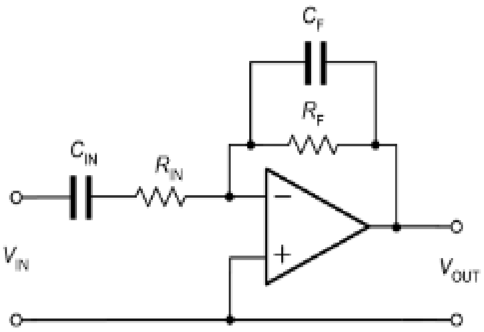


Fig 2.4. Adding capacitors to modify the frequency response of an inverting operational amplifier.

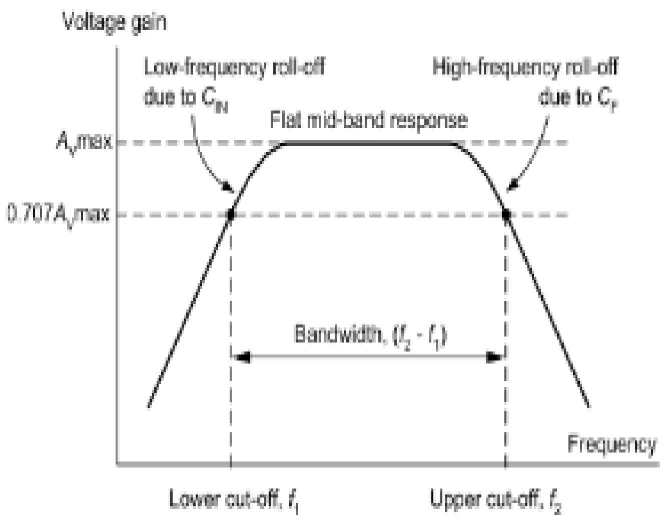


Fig 2.5. Effect of adding capacitors, CIN and CF, to modify the frequency response of an operational amplifier.

Problem:

An inverting operational amplifier is to operate according to the following specification:

Voltage gain = 100

Input resistance (at mid-band) = 10 kΩ

Lower cut-off frequency = 250 Hz

Upper cut-off frequency = 15 kHz

Devise a circuit to satisfy the above specification using an operational amplifier.

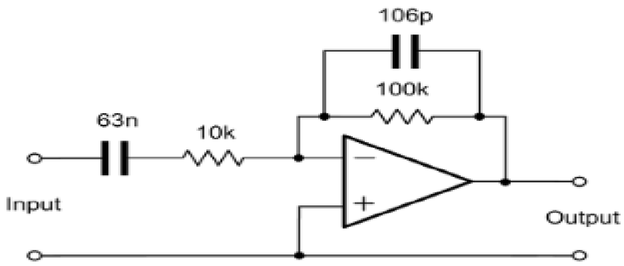
Solution

Circuit on a single operational amplifier configured as an inverting amplifier with capacitors to define the upper and lower cut-off frequencies, as shown in Fig.2.5. The nominal input resistance is the same as the value for R_{IN}. Thus:

$$R_{IN} = 10\ K\omega$$

To determine the value of RF we can make use of the formula for mid-band voltage gain:

$$A_v - \frac{R2}{R1}$$



thus $R2 = A_v \times R1 = 100 \times 10\ \text{k}\Omega = 100\text{k}\Omega$

To determine the value of C_{IN} we will use the formula for the low-frequency cut-off:

$$f_1 = \frac{0.159}{C_{IN}R_{IN}}$$

from which:

$$C_{IN} = \frac{0.159}{f_1 R_{IN}} = \frac{0.159}{250 \times 10 \times 10^3}$$

hence:

$$C_{IN} = \frac{0.159}{2.5 \times 10^6} = 63 \times 10^{-9}\text{F} = 63\ \text{nF}$$

Finally, to determine the value of C_F we will use the formula for high-frequency cut-off:

$$f_2 = \frac{0.159}{C_F R_F}$$

from which:

$$C_F = \frac{0.159}{f_2 R_{IN}} = \frac{0.159}{15 \times 10^3 \times 100 \times 10^3}$$

hence:

$$C_f = \frac{0.159}{1.5 \times 10^9} = 0.106 \times 10^{-9}\text{F} = 106\ \text{pF}$$

For most applications the nearest preferred value (68 nF for C_{IN} and 100 pF for C_F) would be perfectly adequate. The complete circuit of the

Operational amplifier

Operational amplifier circuits:

- As well as their application as a general-purpose amplifying device, operational amplifiers have a number of other uses, including voltage followers, differentiators, integrators, comparators and summing amplifiers.

Voltage followers:

- A voltage follower using an operational amplifier is shown in Fig. 8.11.
- This circuit is essentially an inverting amplifier in which 100% of the output is fed back to the input.
- The result is an amplifier that has a voltage gain of 1 (i.e. unity), a very high input resistance and a very high output resistance. This stage is often referred to as a buffer and is used for matching a high impedance circuit to a low-impedance circuit.
- Typical input and output waveforms for a voltage follower are shown in Fig. 8.12. Notice how the input and output waveforms are both in-phase (they rise and fall together) and that they are identical in amplitude.

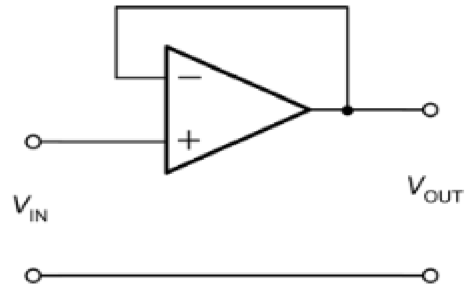


Fig 2.6.A voltage follower

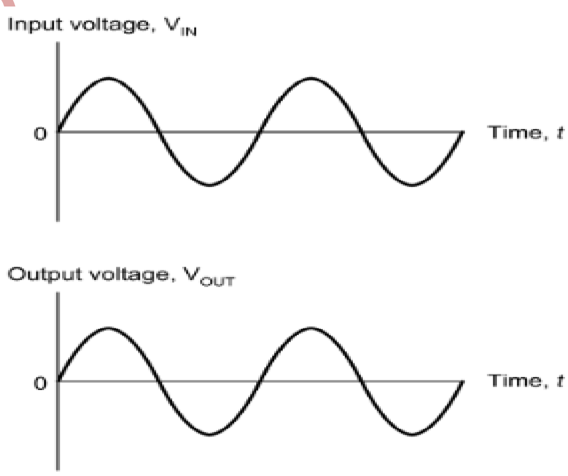


Fig 2.7. Typical input and output waveforms for a voltage follower.

- **Differentiators**
- A differentiator using an operational amplifier is shown in Fig. 2.8.
- A differentiator produces an output voltage that is equivalent to the rate of change of its input.
- This may sound a little complex but it simply means that if the input voltage remains constant (i.e. if it isn't changing) the output also remains constant.
- The faster the input voltage changes the greater will the output be.
- In mathematics this is equivalent to the differential function.
- Typical input and output waveforms for a differentiator are shown in Fig. 2.9.
- Notice how the square wave input is converted to a train of short duration pulses at the output.
- Note also that the output waveform is inverted because the signal has been applied to the inverting input of the operational amplifier.

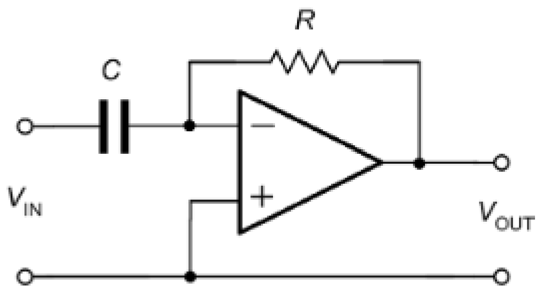


Fig 2.8. A differentiator

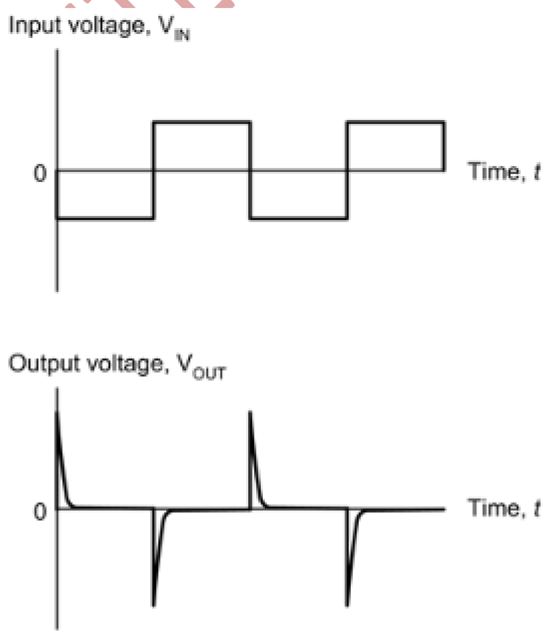


Fig 2.9. Typical input and output waveforms for a differentiator

- **Integrators:**
- An integrator using an operational amplifier is shown in Fig. 2.10.
- This circuit provides the opposite function to that of a differentiator (see earlier) in that its output is equivalent to the area under the graph of the input function rather than its rate of change.
- If the input voltage remains constant (and is other than 0 V) the output voltage will ramp up or down according to the polarity of the input.
- The longer the input voltage remains at a particular value the larger the value of output voltage (of either polarity) will be produced.
- Typical input and output waveforms for an integrator are shown in Fig. 2.11. Notice how the square wave input is converted to a wave that has a triangular shape.
- Once again, note that the output waveform is inverted.

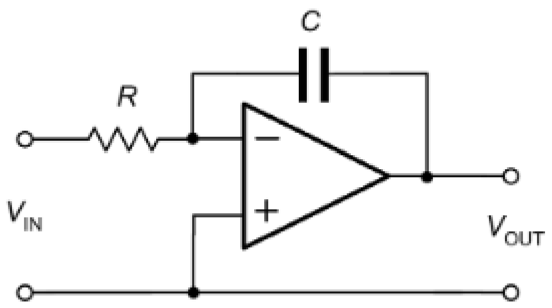


Fig 2.10. An integrator

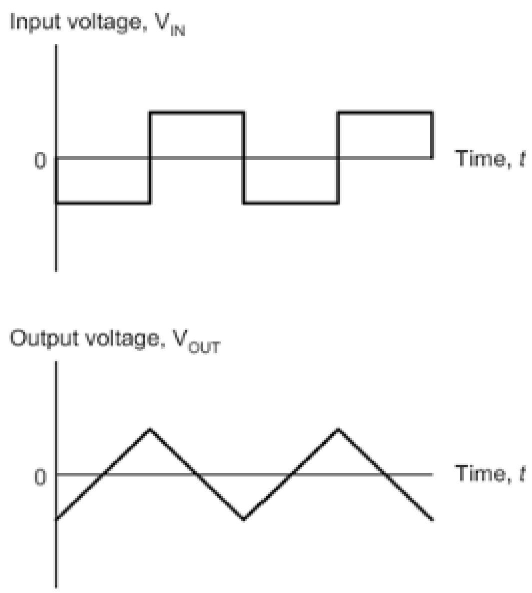


Fig 2.11. Typical input and output waveforms for an integrator.

- **Comparators**
- A comparator using an operational amplifier is shown in Fig. 2.12.
- Since no negative feedback has been applied, this circuit uses the maximum gain of the operational amplifier.
- The output voltage produced by the operational amplifier will thus rise to the maximum possible value (equal to the positive supply rail voltage) whenever the voltage present at the non-inverting input exceeds that present at the inverting input.
- Conversely, the output voltage produced by the operational amplifier will fall to the minimum possible value (equal to the negative supply rail voltage).
- Whenever the voltage present at the inverting input exceeds that present at the non-inverting input.
- Typical input and output waveforms for a comparator are shown in Fig. 2.13.
- Notice how the output is either +15 V or –15 V depending on the relative polarity of the two inputs.
- A typical application for a comparator is that of comparing a signal voltage with a reference voltage.
- The output will go high (or low) in order to signal the result of the comparison.

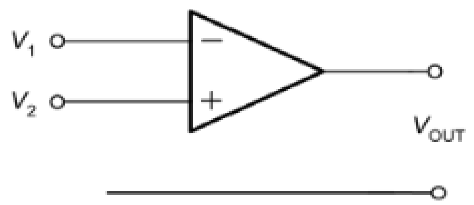


Fig 2.12. a comparator

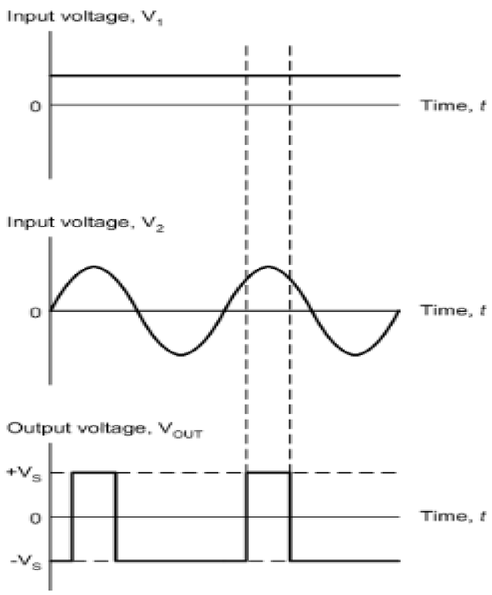


Fig 2.13. Typical input and output waveforms for a comparator.

- **Summing amplifiers**
- A summing amplifier using an operational amplifier is shown in Fig. 2.14.
- This circuit produces an output that is the sum of its two input voltages.
- However, since the operational amplifier is connected in inverting mode, the output voltage is given by: VOUT = – (V1 + V2).
  where V1 and V2 are the input voltages (note that all of the resistors used in the circuit have the same value).
- Typical input and output waveforms for a summing amplifier are shown in Fig. 2.15.
- A typical application is that of 'mixing' two input signals to produce an output voltage that is the sum of the two.

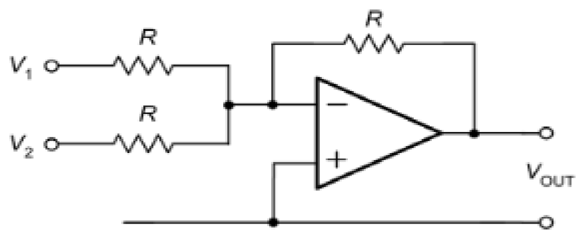


Fig 2.14. A summing amplifier

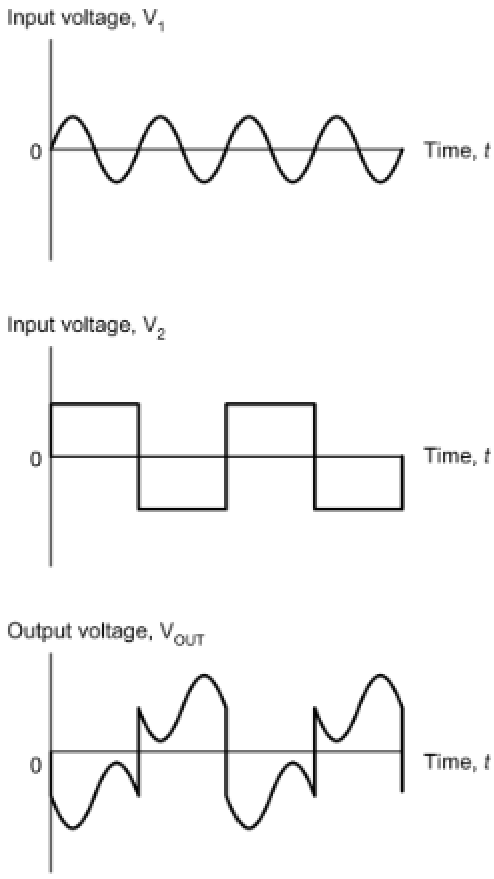
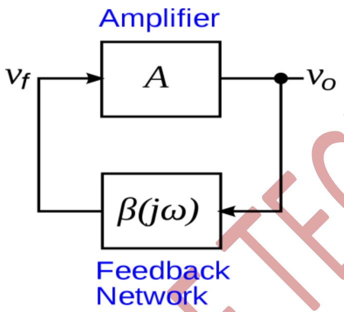


Fig 2.15. Typical input and output waveforms for a summing amplifier.

OSCILLATORS:

- An oscillator is **a mechanical or electronic device that works on the principles of oscillation**: a periodic fluctuation between two things based on changes in energy.
- Computers, clocks, watches, radios, and metal detectors are among the many devices that use oscillators.
- Oscillators are essential components that produce a periodic electronic signal, typically a sine wave or square wave.
- Oscillators **convert DC signal to periodic AC signals which can be used to set frequency, be used for audio applications, or used as a clock signal**.



Difference between amplifier and oscillators:

SL NO.	AMPLIFIERS	OSCILLATORS
1	**Amplifier is an electronic circuit which gives output as amplified form of input.**	**Oscillators are an electronic circuit which gives output without application of input**.
2	The amplifier does not generate any periodic signal.	The oscillators generates periodic signal.
3	Negative feedback.	Positive feedback.

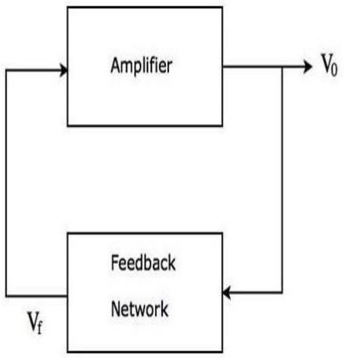
The Barkhausen stability criteria

- The Barkhausen stability criteria is **a mathematical requirement used in electronics to predict whether a linear electronic circuit may oscillate**. Heinrich Georg Barkhausen a German scientist, proposed it.
- if A is the gain of the amplifying element in the circuit and $\beta(j\omega)$ is the feedback path transfer function, so βA is the loop gain around the circuit's feedback loop, the circuit will maintain steady-state oscillations only at frequencies for which:

1. The loop gain is equal to one in absolute magnitude, which means that $\beta A = 1$.

2. The feedback must be positive (i.e. the signal fed back must arrive back in-phase with the signal at the input); **should be 0∘ or 360∘**

Sinusoidal oscillator:



1. A sinusoidal oscillator is an oscillator that generates a periodic signal in the shape of a sinusoidal wave.
2. It is used to convert input energy from a DC source into the periodic signal's AC output energy.
3. Sinusoidal or harmonic oscillators are oscillators that generate an output using a sine waveform.
4. These oscillators are capable of producing output at frequencies between 20 Hz and 1 GHz.
5. This type of periodic signal has a specific frequency and amplitude.
6. A sinusoidal oscillator has two main parts, one is an amplifier and the other one is a feedback network.
7. Though, it has a feedback network the loop gain of a sinusoidal oscillator is greater than or equal to unity.
8. The total phase shift around the loop in a sinusoidal oscillator is either 0∘or 360∘.
9. There are a few types of sinusoidal oscillators:

- Tuned circuit oscillator:
- RC oscillators:
- Crystal oscillators:
- Negative-resistance oscillator

- **Ladder network oscillator:**
- A simple phase-shift oscillator based on a three-stage C–R ladder network is shown in Fig. 2.16.
- TR1 operates as a conventional common-emitter amplifier stage with R1 and R2 providing base bias potential and R3 and C1 providing emitter stabilization.
- The total phase shift provided by the C–R ladder network (connected between collector and base) is 180° at the frequency of oscillation.
- The transistor provides the other 180° phase shift in order to realize an overall phase shift of 360° or 0° (note that these are the same).
- The frequency of oscillation of the circuit shown in Fig.2.16 is given by:

$$f = \frac{1}{2\pi \times \sqrt{6}CR}$$

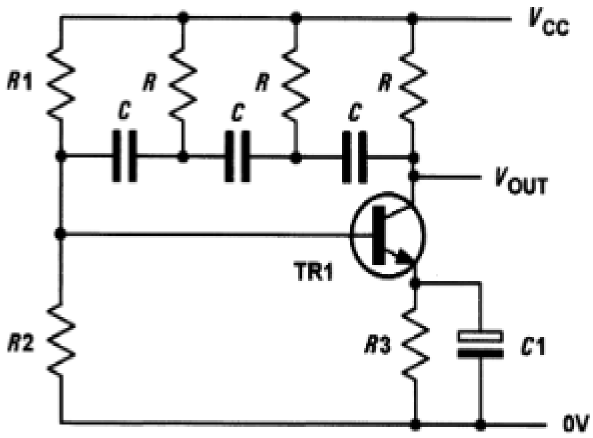


Fig 2.16. Sine wave oscillator based on a three-stage C–R ladder network

- The loss associated with the ladder network is 29, thus the amplifier must provide a gain of at least 29 in order for the circuit to oscillate.
- In practice this is easily achieved with a single transistor.

PROBLEM :

Determine the frequency of oscillation of a three-stage ladder network oscillator in which C = 10 nF and R = 10 kΩ.

Solution

Using

$$f = \frac{1}{2\pi \times \sqrt{6}CR}$$

gives

$$f = \frac{1}{6.28 \times 2.45 \times 10 \times 10^{-9} \times 10 \times 10^{3}}$$

from which

$$f = \frac{1}{6.28 \times 2.45 \times 10^{-4}} = \frac{10^{4}}{15.386} = 647 \text{ Hz}$$

- **Wien bridge oscillator:**
- An alternative approach to providing the phase shift required is the use of a Wien bridge network (Fig. 2.17).
- Like the C–R ladder, this network provides a phase shift which varies with frequency.
- The input signal is applied to A and B while the output is taken from C and D.
- At one particular frequency, the phase shift produced by the network will be exactly zero (i.e. the input and output signals will be in-phase).
- If we connect the network to an amplifier producing 0° phase shift which has sufficient gain to overcome the losses of the Wien bridge, oscillation will result.
- The minimum amplifier gain required to sustain oscillation is given by:

$$A_v = 1 + \frac{C1}{C2} + \frac{R2}{R1}$$

- In most cases, C1 = C 2 and R1 = R2, hence the minimum amplifier gain will be 3.

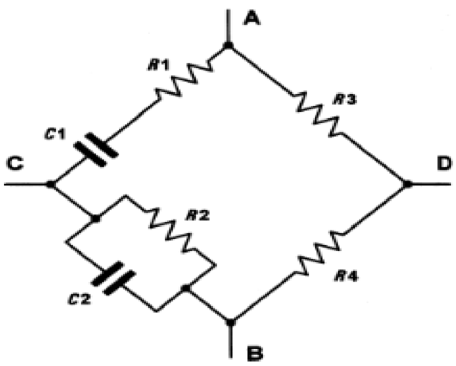


Fig 2.17. A Wien bridge network

The frequency at which the phase shift will be zero is given by:

$$f = \frac{1}{2\pi \times \sqrt{C1C2R1R2}}$$

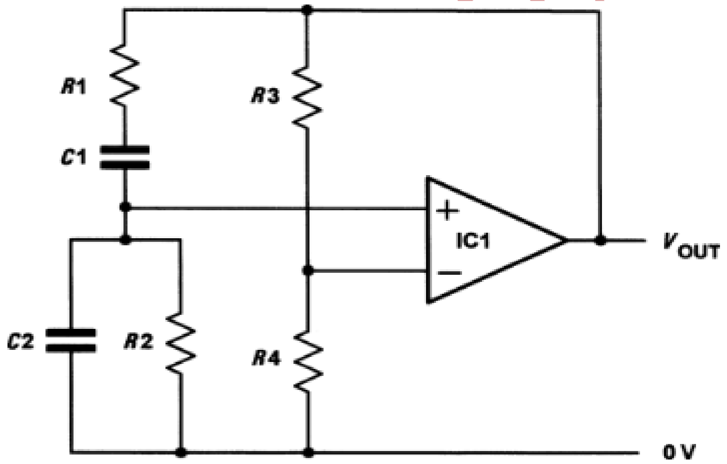
When $R1 = R2$ and $C1 = C2$ the frequency at which the phase shift will be zero will be given by:

$$f = \frac{1}{2\pi \times \sqrt{C^2R^2}} = \frac{1}{2\pi CR}$$

where $R = R1 = R2$ and $C = C1 = C2$.

PROBLEM:

Fig. 9.4 shows the circuit of a Wien bridge oscillator based on an operational amplifier. If C1 = C 2 = 100 nF, determine the output frequencies produced by this arrangement (a) when R1 = R2 = 1 kΩ and (b) when R1 = R2 = 6 kΩ.



Solution

(a) When $R1 = R2 = 1\ \text{k}\Omega$

$$f = \frac{1}{2\pi CR}$$

where $R = R1 = R1$ and $C = C1 = C2$.

Thus

$$f = \frac{1}{6.28 \times 100 \times 10^{-9} \times 1 \times 10^{3}}$$

$$f = \frac{10^4}{6.28} = 1.59\ \text{kHz}$$

(b) When $R1 = R1 = 6\ k\Omega$

$$f = \frac{1}{2\pi CR}$$

where $R = R1 = R1$ and $C = C1 = C2$.

Thus

$$f = \frac{1}{6.28 \times 100 \times 10^{-9} \times 6 \times 10^{3}}$$

$$f = \frac{10^4}{37.68} = 265\ \text{Hz}$$

- **Multivibrators:**
- There are many occasions when we require a square wave output from an oscillator rather than a sine wave output.
- Multivibrators are a family of oscillator circuits that produce output waveforms consisting of one or more rectangular pulses.
- The term ‘multivibrator’ simply originates from the fact that this type of waveform is rich in harmonics (i.e. ‘multiple vibrations’).
- Multivibrators use regenerative (i.e. positive) feedback; the active devices present within the oscillator circuit being operated as switches, being alternately cut-off and driven into saturation.

  The principal types of multivibrator are:

  (a) astable multivibrators that provide a continuous train of pulses (these are sometimes also referred to as free-running multivibrators);

  (b) monostable multivibrators that produce a single output pulse (they have one stable state and are thus sometimes also referred to as ‘one-shot’);

  (c) bistable multivibrators that have two stable states and require a trigger pulse or control signal to change from one state to another.

- **Single-stage astable oscillator:**
- A simple form of astable oscillator that produces a square wave output can be built using just one operational amplifier, as shown in Fig. 9.10.
- The circuit employs positive feedback with the output fed back to the non-inverting input via the potential divider formed by R1 and R2.

- This circuit can make a very simple square wave source with a frequency that can be made adjustable by replacing R with a variable or preset resistor.
- Assume that C is initially uncharged and the voltage at the inverting input is slightly less than the voltage at the non-inverting input.
- The output voltage will rise rapidly to +VCC and the voltage at the inverting input will begin to rise exponentially as capacitor C charges through R.
- Eventually the voltage at the inverting input will have reached a value that causes the voltage at the inverting input to exceed that present at the non-inverting input.
- At this point, the output voltage will rapidly fall to −VCC. Capacitor C will then start to charge in the other direction and the voltage at the inverting input will begin to fall exponentially.
- Eventually, the voltage at the inverting input will have reached a value that causes the voltage at the inverting input to be less than that present at the non-inverting input.
- At this point, the output voltage will rise rapidly to +VCC once again and the cycle will continue indefinitely.
- The upper threshold voltage (i.e. the maximum positive value for the voltage at the inverting input) will be given by:

$$V_{UT} = V_{CC} \times \left(\frac{R2}{R1+R2}\right)$$

- The lower threshold voltage (i.e. the maximum negative value for the voltage at the inverting input) will be given by:

$$V_{LT} = -V_{CC} \times \left(\frac{R2}{R1+R2}\right)$$

- Finally, the time for one complete cycle of the output waveform produced by the astable oscillator is given by:

$$T = 2CR\ln\left(1+2\left(\frac{R2}{R1}\right)\right)$$

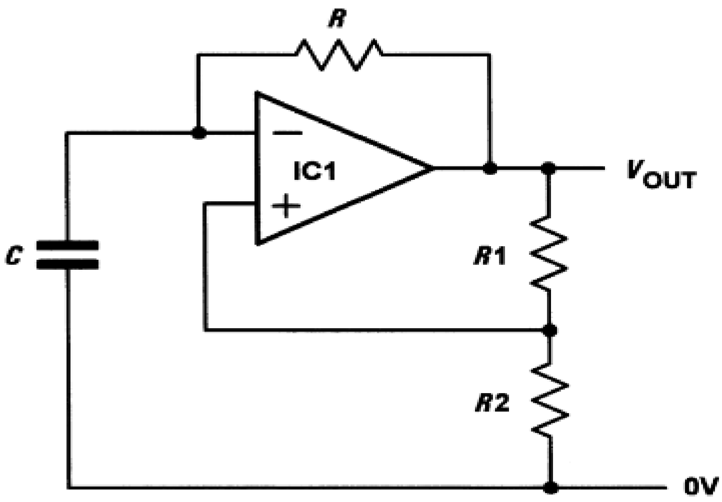


Fig 2.18. Single-stage astable oscillator using an operational amplifier.

- **Crystal controlled oscillators:**
- A requirement of some oscillators is that they accurately maintain an exact frequency of oscillation.
- In such cases, a quartz crystal can be used as the frequency determining element.
- The quartz crystal (a thin slice of quartz in a hermetically sealed enclosure, see Fig. 2.19)
- Vibrates whenever a potential difference is applied across its faces (this phenomenon is known as the piezoelectric effect).
- The frequency of oscillation is determined by the crystal's 'cut' and physical size.
- Most quartz crystals can be expected to stabilize the frequency of oscillation of a circuit to within a few parts in a million.
- Crystals can be manufactured for operation in fundamental mode over a frequency range extending from 100 kHz to around 20 MHz and for overtone operation from 20 MHz to well over 100 MHz.

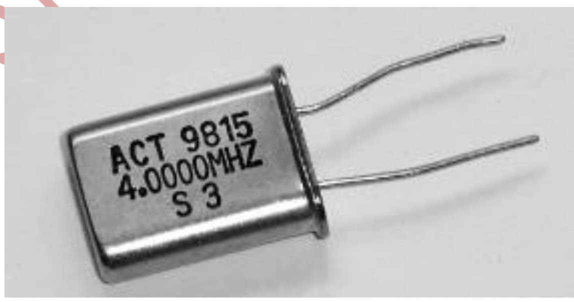


FIG 2.19. CRYSTAL OSCILLATOR.